A 3×3 Counterexample to the Indefinite Reverse-Order Rank-Free Implication

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Status. Candidate counterexample, likely valid, for human review.

1 Source question

Kamaraj, Johnson, and Satheesh study Moore–Penrose inverses between indefinite inner product spaces. Their Theorem 3.16 proves, under a displayed rank equality, that the reverse order law $(AB)^{[\dagger]} = B^{[\dagger]}A^{[\dagger]}$ is equivalent to the four conditions of their Theorem 3.5. Section 4 asks whether the theorem remains true without the rank equality, or whether a counterexample exists [1, Section 4, PDF p. 10].

4. An Open problem

We can observe from Theorem 3.16 that $(AB)^{[\dagger]} = B^{[\dagger]}A^{[\dagger]}$ is equivalent to any one of the conditions given in Theorem 3.5 by assuming the rank equality

$$rank \begin{pmatrix} B^{[*]}A^{[*]} & B^{[*]}B \\ AA^{[*]} & AB \end{pmatrix} = rank \begin{pmatrix} A^{[*]} & B \end{pmatrix}.$$

But in the Euclidean case such a rank equality assumption is not required. Thus it is an open question for giving proof for Theorem 3.16 without assuming the rank equality, or finding a counter example.

More explicitly, the assumption in question is

$$\operatorname{rank} \begin{pmatrix} B^{[*]}A^{[*]} & B^{[*]}B \\ AA^{[*]} & AB \end{pmatrix} = \operatorname{rank} \begin{pmatrix} A^{[*]} & B \end{pmatrix}.$$

The requested rank-free equivalence would in particular assert that the reverse order law forces condition (i) of Theorem 3.5: the matrix $A^{[*]}ABB^{[*]}$ is range Hermitian.

2 Definitions

We use the same weight J on all three copies of $\mathbb{C}^3$. Thus

$$X^{[*]} = J^{-1}X^*J = JX^*J.$$

The indefinite Moore–Penrose inverse $X^{[\dagger]}$, when it exists, is the unique matrix Y satisfying

$$XYX = X, \qquad YXY = Y, \qquad (XY)^{[*]} = XY, \qquad (YX)^{[*]} = YX.$$

A square matrix T is range Hermitian when $\mathrm{Ran}(T) = \mathrm{Ran}(T^{[*]})$.

3 Proof intuition

The Euclidean reverse-order theorem cannot see the following indefinite configuration. There are distinct nonisotropic vectors u, v of the same positive J-norm whose mutual J-inner product attains the formal Cauchy–Schwarz equality. Their span is a degenerate plane. The J-orthogonal rank-one projections P and Q onto these two lines do not commute, but both ordered products are idempotent. Taking $A = Q$ and $B = P$, the product in one order is the indefinite Moore–Penrose inverse of the product in the other order. This gives the reverse order law, while the distinct ranges of the two products immediately violate range Hermiticity.

4 Counterexample

Theorem 1. *The rank assumption in Theorem 3.16 of [1] cannot be removed. In fact, there are real* 3×3 *matrices* A, B*, with the common indefinite Hermitian weight* $J = \mathrm{diag}(1, 1, -1)$*, such that*

$$A^{[\dagger]},\ B^{[\dagger]},\ (AB)^{[\dagger]} \ \textit{exist}, \qquad (AB)^{[\dagger]} = B^{[\dagger]}A^{[\dagger]},$$

but none of the equivalent conditions of Theorem 3.5 holds. The displayed rank equality in Theorem 3.16 has left-hand rank 1 *and right-hand rank* 2*.*

Proof. Set

$$u = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad B = P := uu^*J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and

$$A = Q := vv^*J = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{pmatrix}.$$

The three scalar products relevant to the construction are

$$u^*Ju = 1, \qquad v^*Jv = 1, \qquad u^*Jv = v^*Ju = 1.$$

Consequently

$$P^2 = P = P^{[*]}, \qquad Q^2 = Q = Q^{[*]}.$$

Each of P, Q therefore satisfies the four Penrose equations with itself, so

$$B^{[\dagger]} = B = P, \qquad A^{[\dagger]} = A = Q.$$

The two ordered products are

$$D := AB = QP = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad X := BA = PQ = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since P, Q are J-Hermitian, $D^{[*]} = X$. Direct multiplication gives

$$DX = Q, \qquad XD = P, \qquad DXD = D, \qquad XDX = X.$$

The products $DX = Q$ and $XD = P$ are J-Hermitian. Hence X satisfies all four defining equations for $D^{[\dagger]}$, and therefore

$$(AB)^{[\dagger]} = D^{[\dagger]} = X = PQ = B^{[\dagger]}A^{[\dagger]}.$$

Condition (i) of Theorem 3.5 nevertheless fails. Indeed,

$$A^{[*]}ABB^{[*]} = Q^2P^2 = QP = D, \qquad (A^{[*]}ABB^{[*]})^{[*]} = D^{[*]} = PQ = X.$$

But

$$\operatorname{Ran}(D) = \operatorname{span}\{v\}, \qquad \operatorname{Ran}(D^{[*]}) = \operatorname{Ran}(X) = \operatorname{span}\{u\},$$

and these lines are distinct. Thus $A^{[*]}ABB^{[*]}$ is not range Hermitian. Since the source proves conditions (i)–(iv) equivalent without the extra rank assumption, all four fail. For a completely entrywise check of condition (iv), for example,

$$BB^{[\dagger]}A^{[*]}AB = PQP = P \neq QP = A^{[*]}AB.$$

Finally, the two ranks in the omitted hypothesis are transparent. Its block matrix is

$$\begin{pmatrix} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & -1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 1 & 0 & 0 \end{pmatrix},$$

which has rank 1, whereas

$$\begin{pmatrix} A^{[*]} & B \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 0 \end{pmatrix}$$

has rank 2. This is precisely a counterexample of the type requested in Section 4. $\square$

5 Verification and limitations

The exact symbolic checker in `code/verify_counterexample.py` verifies the weighted adjoint and idempotency identities, all four Penrose equations, the reverse order law, failure of conditions (i) and (iv), and the ranks 1 and 2. It runs with:

```
conda run --no-capture-output -n sandbox python \
  code/verify_counterexample.py
```

All calculations use exact integers and rationals. The script is an audit, not a substitute for the entrywise proof above.

The counterexample answers the source's yes/no question negatively with real 3×3 matrices. It does not provide a replacement rank-free characterization and does not claim that dimension three is minimal.

6 Novelty check

The bounded search on 9 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes, followed by web searches for arXiv:2108.05873, the exact paper title, the exact rank-equality phrase, "indefinite reverse order law counterexample," and the core expression $A^{[*]}ABB^{[*]}$. It found the arXiv source and the 2023 published version, but no later work explicitly answering Section 4. This supports, but cannot certify, novelty.

Human-review recommendation. Verify alignment with the source's weighted-adjoint convention. After that check, the counterexample is a short direct negative answer.

References

[1] K. Kamaraj, P. Sam Johnson, and Athira Satheesh K., *Reverse Order Law for Generalized Inverses with Indefinite Hermitian Weights*, arXiv:2108.05873 (2021); Filomat **37** (2023), 699–709, https://arxiv.org/abs/2108.05873.